

Mini-Project Report: Computer Software Setup and Security

ProjectTitle:ComputerSoftware SetupandSecurity(Drivers,OfficeApplications,and Antivirus Protection)

Completion Date: October 03, 2026

Computer Name (Host Name): DESKTOP-P69IP4H

1. Student Information and Computer Specifications

Parameter	Details / Specification
Student Full Name	Zhumabay Nurdaulet
Academic Group	SE 1-24
Computer Name	DESKTOP-P69IP4H
Processor (CPU)	AMD Ryzen 5 7535HS with Radeon Graphics (6 cores, 12 logical processors)
RAM	16 GB DDR5
Graphics Cards (GPU)	1. Integrated: AMD Radeon(TM) Graphics 2. Discrete: NVIDIA GeForce RTX 3050 Laptop GPU
Storage	NVMe SSD 512 GB
Operating System	Microsoft Windows 11 Home (64-bit)

2. WEEK 1 — Topic 1.4: Installing and Configuring

Drivers

2.1 Task 1: Computer Diagnosis

Before installing or updating software, a comprehensive hardware audit was performed using Windows **Device Manager**. The system check confirmed that all hardware devices were recognized and operating without hardware conflicts or driver errors. No yellow exclamation marks, missing drivers, or unknown devices were detected.

Component	Model / Name	Driver Status	Problem / Action Taken
CPU	AMD Ryzen 5 7535HS with Radeon Graphics	Working correctly	No issues. AMD Chipset drivers updated.
RAM	16 GB DDR5	Working correctly	No issues. Managed by system BIOS/UEFI firmware.
Graphics Card (Discrete)	NVIDIA GeForce RTX 3050 Laptop GPU	Reinstalled / Updated	No initial problems. Clean reinstallation of GeForce Game Ready Driver v617.14 performed.
Graphics Card (Integrated)	AMD Radeon(TM) Graphics	Working correctly	No initial problems. Updated alongside AMD Chipset Drivers.
Network Adapter	Realtek / AMD Wi-Fi & Ethernet Adapters	Working correctly	No issues. Network connections stable.
Audio Device	Realtek High Definition Audio / NVIDIA HD Audio	Working correctly	No issues. Audio input and output working properly.
Printer	Virtual Printer (Microsoft Print to	Ready	No physical printer attached; spooler

Component	Model / Name	Driver Status	Problem / Action Taken
	PDF)		service active.

2.2 Task 2 & 3: Driver Audit and Installation Process

Initial Status: As observed in Device Manager, the computer had no driver errors or missing devices prior to setup. To satisfy practical project requirements and ensure optimal system performance, routine maintenance and updates were carried out for two main hardware components:

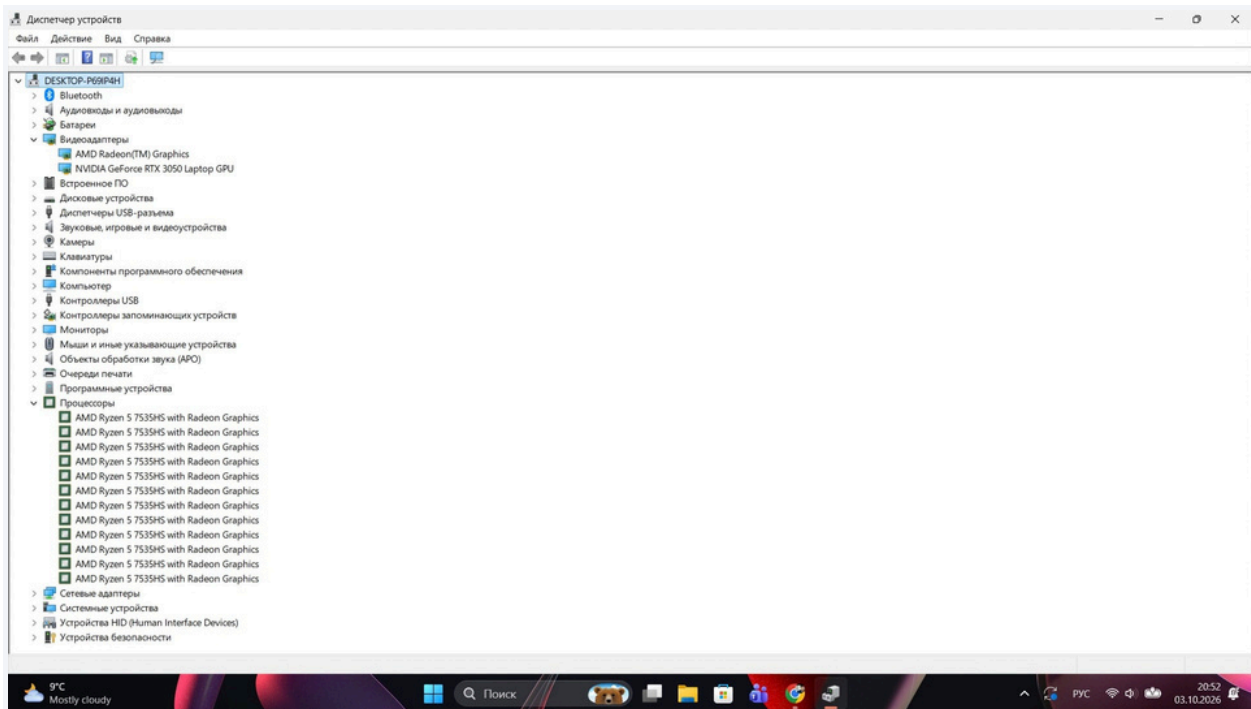
1. Discrete GPU Driver Reinstallation (NVIDIA GeForce RTX 3050 Laptop GPU):

- **Manufacturer:** NVIDIA Corporation
- **Source:** Official NVIDIA App / Official NVIDIA Website
- **Driver Version:** GeForce Game Ready Driver v617.14 (Release Date: Sep 22, 2026)
- **Process:** Downloaded the official package, executed a clean driver reinstallation, updating PhysX System Software (v9.26.0703) and HD Audio Driver (v1.4.6.3).

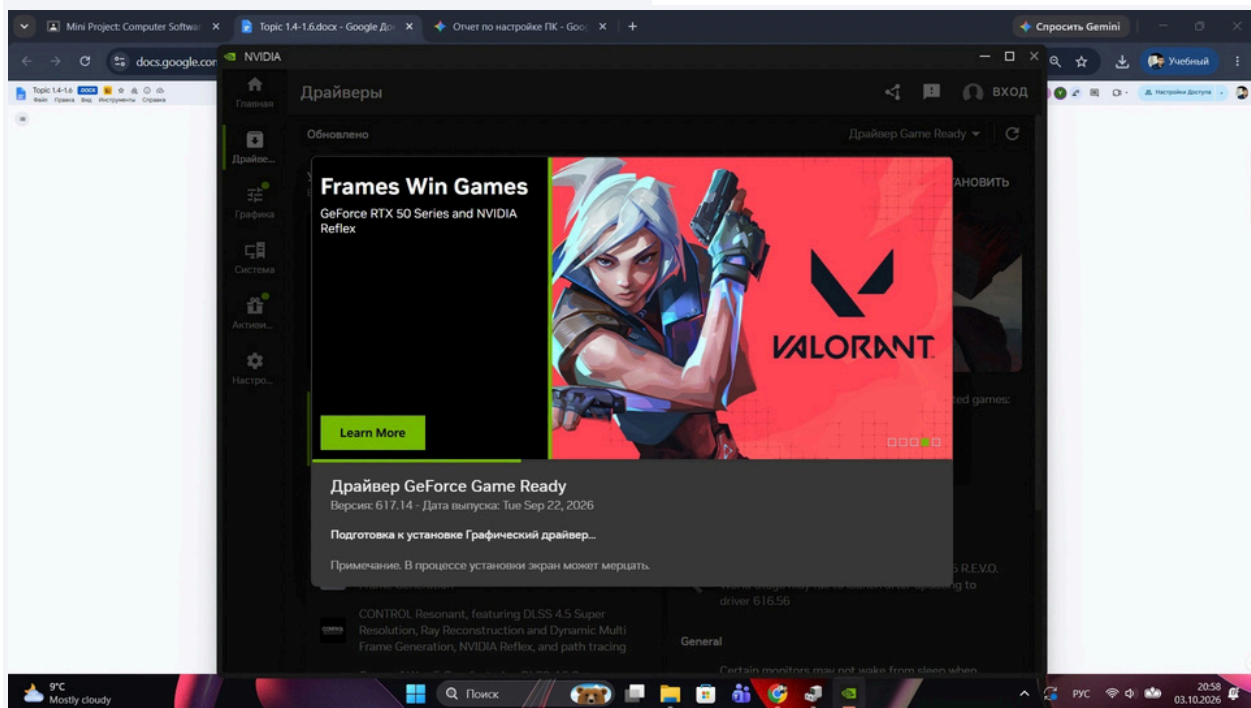
2. AMD System Chipset Drivers Update:

- **Manufacturer:** Advanced Micro Devices (AMD)
- **Source:** Official AMD Software: Adrenalin Edition (v26.2.2) / AMD Install Manager
- **Process:** Initiated chipset driver installation via AMD Install Manager. Upon completion, a system restart was performed to finalize driver registration.

[Screenshot Evidence 1 — devicemanager_2.jpg]



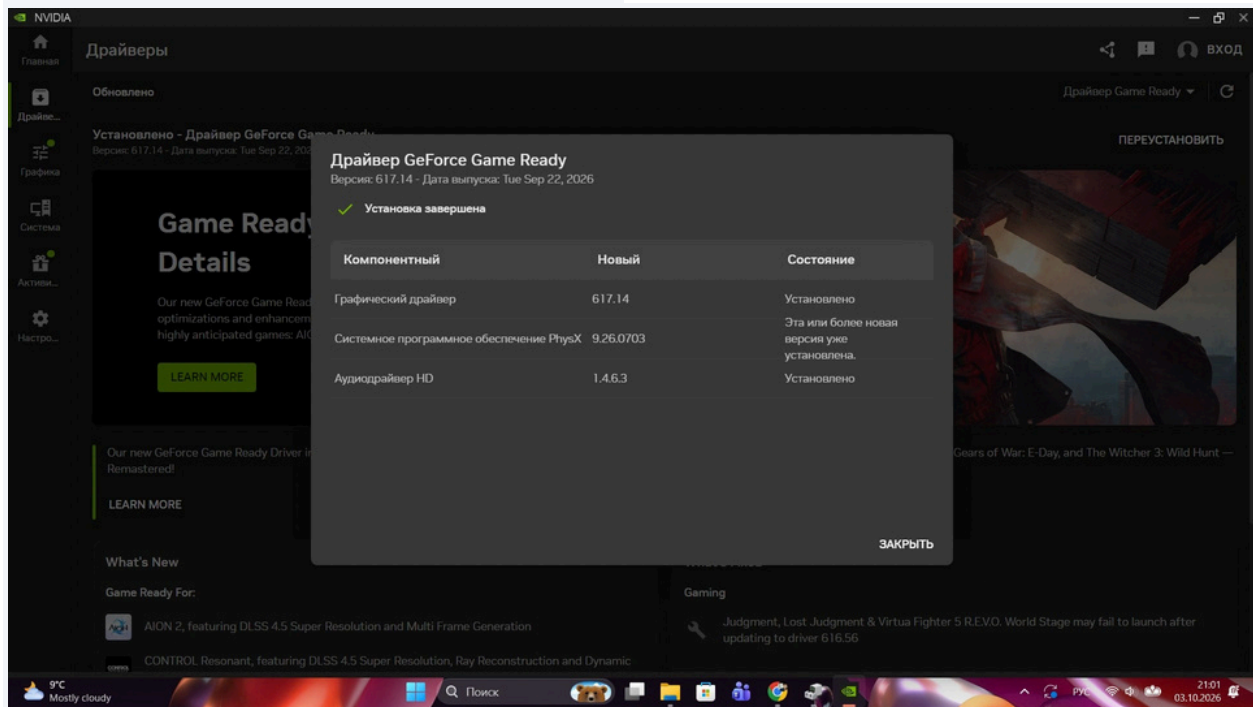
Description: Initial inspection of Windows Device Manager (20:52:02, 03.10.2026). All device trees are expanded. No unknown devices ("Unknown Device") or warning icons are present.
[Screenshot Evidence 2 — GPUdriverinstalling.jpg]



Description: Launching the installation of NVIDIA GeForce Game Ready driver version 617.14 (20:58:14) in NVIDIA App. Status: "Preparing to install Graphics Driver...".
[Screenshot Evidence 3 — gpudriverinstalled.jpg]

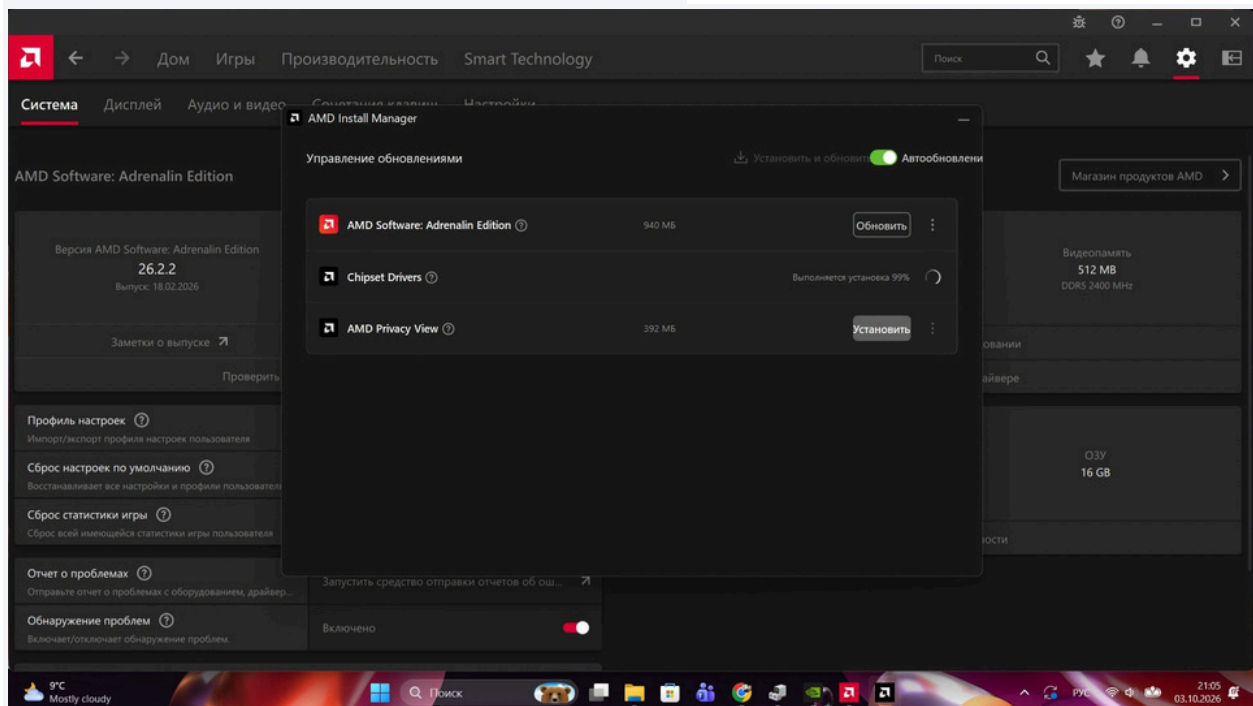
Description: Successful completion of NVIDIA driver installation (21:01:31). Component status: Graphics Driver 617.14 (Installed), PhysX System Software 9.26.0703 (Installed), HD Audio Driver 1.4.6.3 (Installed).

[Screenshot Evidence 4 — DMaftergpubdriver.jpg]

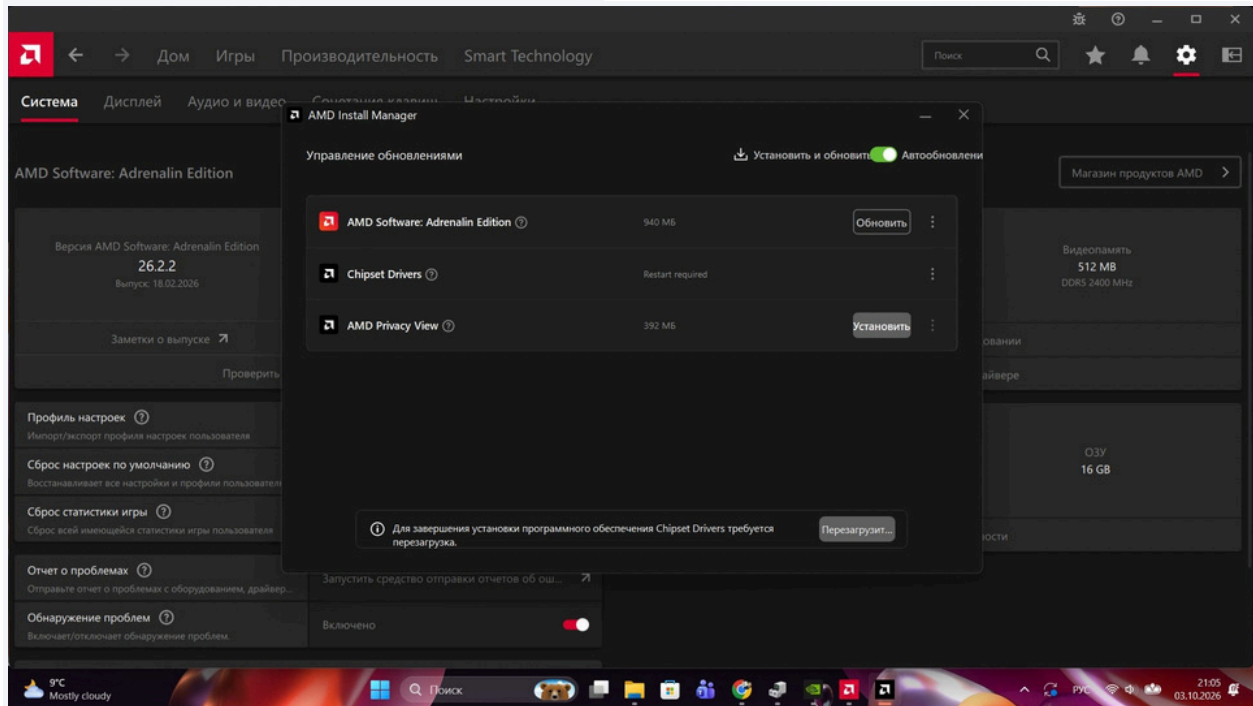


Description: Post-installation check in Device Manager following the NVIDIA GPU update (21:02:45). Display adapters initialized correctly without errors.

[Screenshot Evidence 5 — chipsetdriverinstalling.jpg]

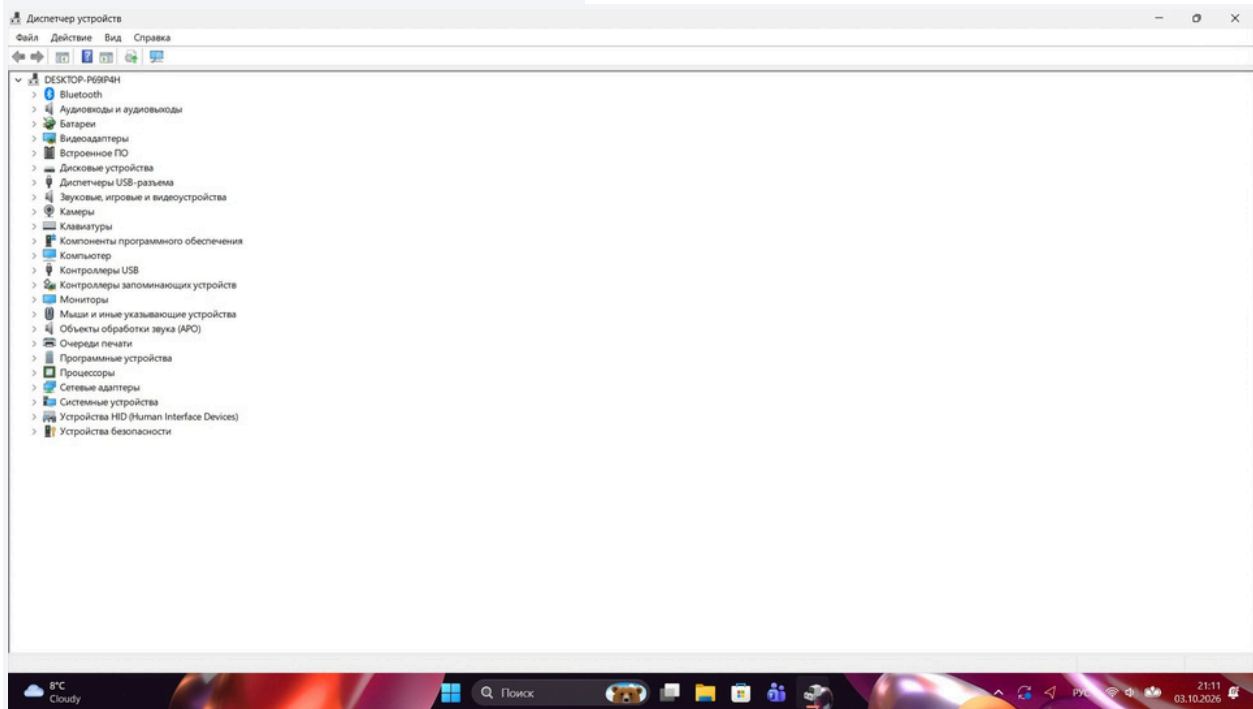


Description: AMD Chipset Drivers installation progress in AMD Install Manager / AMD Software: Adrenalin Edition (21:05:07). Progress: 99%.
[Screenshot Evidence 6 — chipsetinstalled.jpg]



Description: Notification of completed AMD Chipset Drivers installation (21:05:33). Prompt: "Restart required to complete the installation of Chipset Drivers".

[Screenshot Evidence 7 — afterrebootDM.jpg]



Description: Final verification in Device Manager after system restart (21:11:03). Device tree is clean; AMD Ryzen 5 7535HS processors, AMD Radeon, and NVIDIA RTX 3050 graphics operate smoothly without conflicts.

2.3 Short Report — Theoretical Answers

- **What is a device driver?**

A device driver is a specialized software program that allows the operating system to communicate with hardware components by translating high-level OS commands into low-level device instructions.

- **Why are drivers necessary?**

Drivers enable the operating system to utilize hardware features (e.g., 3D graphics acceleration, Wi-Fi connectivity, audio processing, and power management).

- **What can happen if a driver is missing?**

Without drivers, hardware may fail to work entirely or operate in a basic fallback mode with reduced performance, incorrect resolution, lack of audio, or system instability (e.g., Blue Screen of Death / BSOD).

- **Why should drivers be downloaded from official sources?**

Downloading drivers exclusively from official manufacturers (e.g., NVIDIA, AMD, Microsoft) prevents malware infection, ensures system stability, and avoids incompatible or corrupted driver binaries.

- **How can you check whether a driver is installed correctly?**

Open Device Manager and verify that no yellow exclamation marks or error icons appear next to the device name, and ensure its Device Properties state "This device is working properly."

3. WEEK 2 — Topic 1.5: Installing and Configuring Office Applications

3.1 Task 4: Select Office Software

For office application deployment, **WPS Office** was selected as an all-in-one productivity suite compatible with modern Windows operating systems.

Parameter	Details
Software Name	WPS Office

Parameter	Details
Version	12.1.0.25863
Developer	Zhuhai Kingsoft Office Software Co., Ltd.
OS Compatibility	Windows 7 / 8 / 8.1 / 10 / 11 (32-bit & 64-bit)
License Type	Freemium / Proprietary (Free Personal License)
Installation Source	Official WPS Website (explore.wps.com)

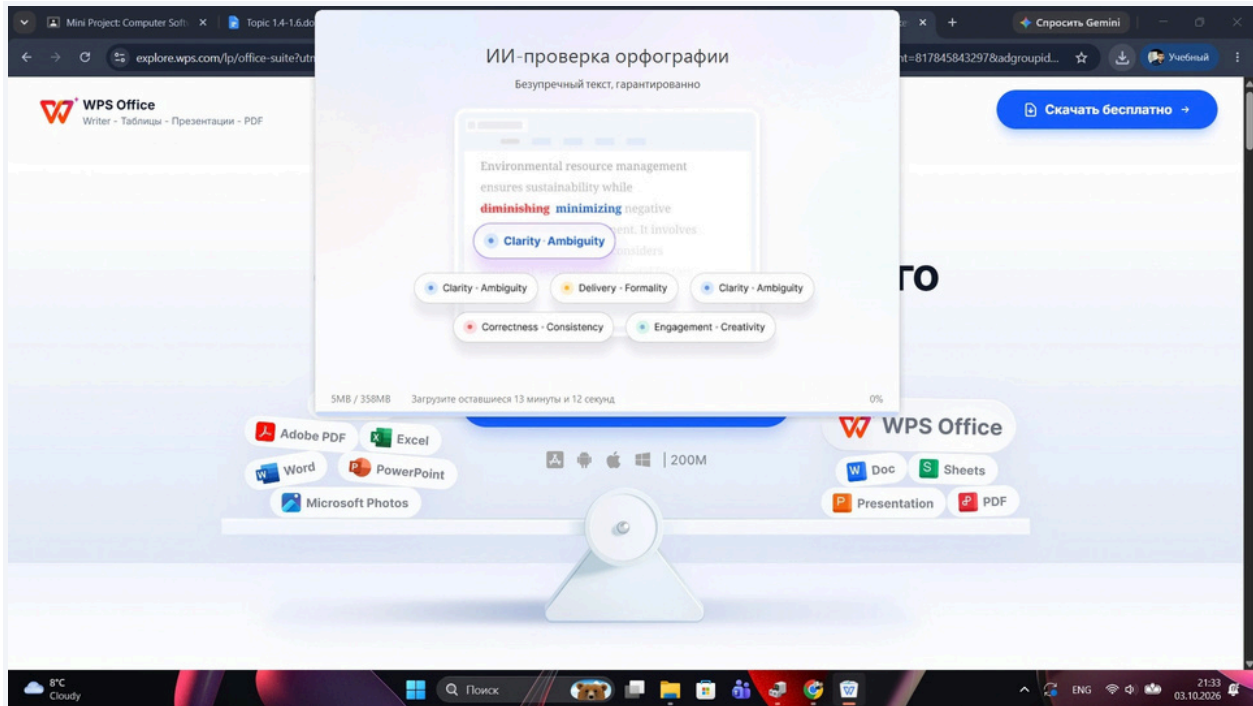
System Requirements for WPS Office:

- **Operating System:** Windows 7/8 / 8.1 / 10 / 11
- **Processor (CPU):** Dual-core processor or higher
- **RAM:** 2 GB or more
- **Free Disk Space:** 4GB or more recommended
- **Architecture Support:** Both 32-bit and 64-bit Windows supported

3.2 Task 5: Installation Process

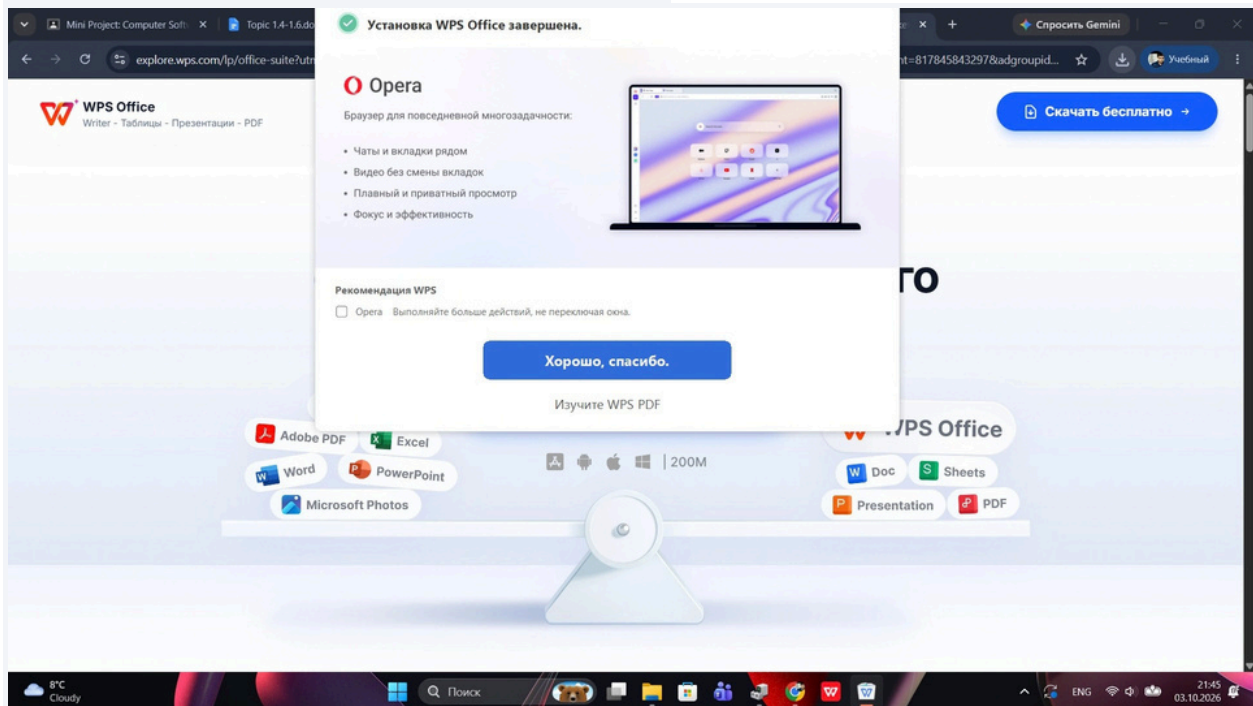
Installation Workflow: The installation was executed via the official WPS Office one-click express installer. No custom component selection was required because the installer automatically sets up all integrated productivity modules (WPS Writer, WPS Spreadsheets, WPS Presentation, and WPS PDF) in a single unified workspace.

[Screenshot Evidence 8 — Снимок экрана (190).jpg]



Description: Downloading WPS Office setup package and progress bar of the automated installation (5 MB / 358 MB) at 21:33:00.

[Screenshot Evidence 9 — Снимок экрана (191).jpg]



Description: Completion screen of WPS Office installation ("Установка WPS Office завершена") at 21:45:00.

3.3 Task 6: Configure Applications

- **Word Processor (WPS Writer):** Language set to English / Russian; Default font set to Arial 11 pt; Line spacing set to 1.15; AutoSave / AutoRecovery enabled (interval: 5 minutes); Default save format: `.docx`.
- **Spreadsheet Application (WPS Spreadsheets):** Date format set to DD.MM.YYYY; Number format configured with standard thousand separators; Default save format: `.xlsx`.
- **Presentation Application (WPS Presentation):** Aspect ratio set to 16:9 Widescreen; Default save format: `.pptx`; Auto-recovery enabled.

3.4 Task 7: Create Test Files

1. **Text Document (`Document.docx`):** Contains title, student info, creation date, security essay, structured parameters table, and security graphic.
2. **Spreadsheet (`Spreadsheet.xlsx`):** Contains a student grade sheet with 10 rows. Calculates average grades automatically using the `=AVERAGE(B2:D2)` formula.
3. **Presentation (`Presentation.pptx`):** A 5-slide slide deck titled "Safe Computer Configuration" covering passwords, updates, drivers, antivirus, and backup strategies.

4. WEEK 3 — Topic 1.6: Installing and Configuring Antivirus Software

3.1 Task 8: Security Check Before Installation

Prior to installing third-party security software, an initial security audit was performed on the workstation (DESKTOP-P69IP4H) to document the pre-installation security baseline.

Security Parameter	Current Status / Baseline Record
Operating System Security Status	Windows 11 Home (64-bit); Core Isolation and System Guard active
Antivirus Status	Windows Security (Windows Defender) active prior to third-party installation

Firewall Status

Windows Defender Firewall operational across Domain, Private, and Public network profiles

Security Updates

System up to date via Windows Update; latest security patches applied

Last Security Scan

Routine Quick Scan completed with 0 threats or malicious objects detected

3.2 Task 9: Antivirus Installation Process (Malwarebytes)

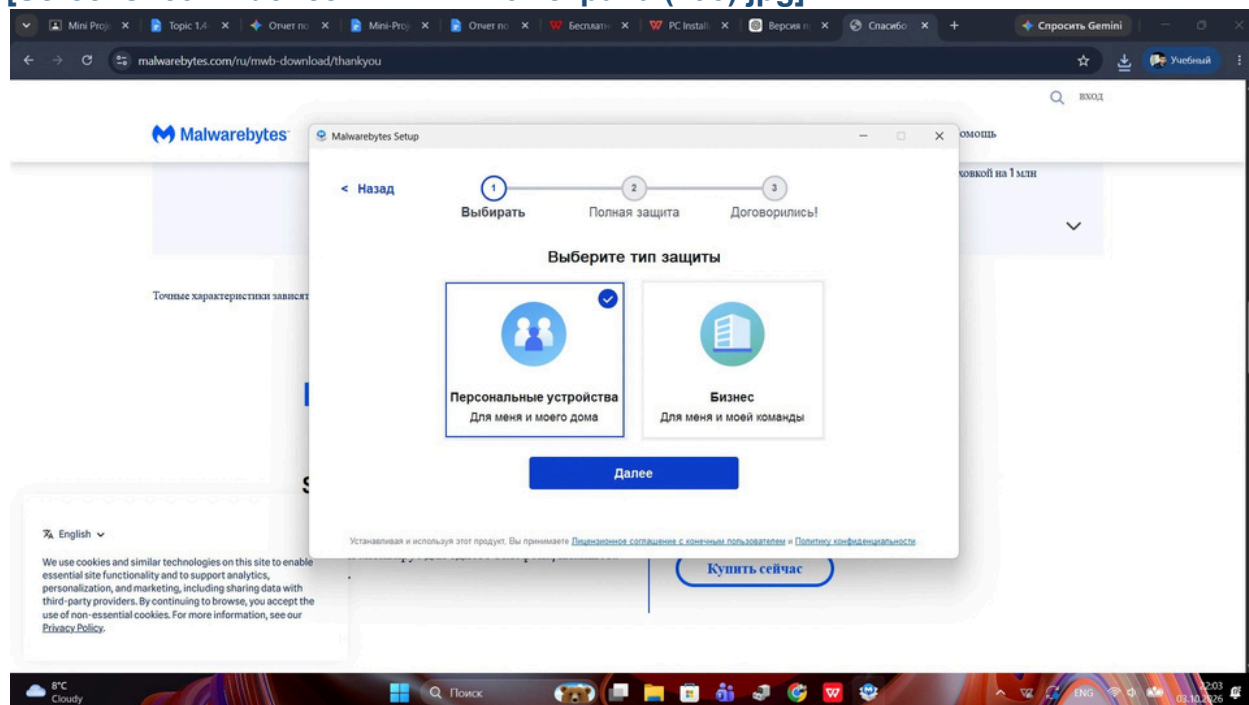
Selected Solution: Malwarebytes (14-day Premium Trial) was selected as an approved security and antimalware solution to provide multi-layered threat detection alongside built-in OS protection.

Installation Workflow Note: Modern Malwarebytes setup utilizes an automated express installer. The installer automatically registers real-time protection drivers and background system services without requiring manual configuration of technical parameters during installation. The user selects the installation context ("Personal Devices") and accepts or declines optional web protection components.

Step-by-Step Installation Workflow:

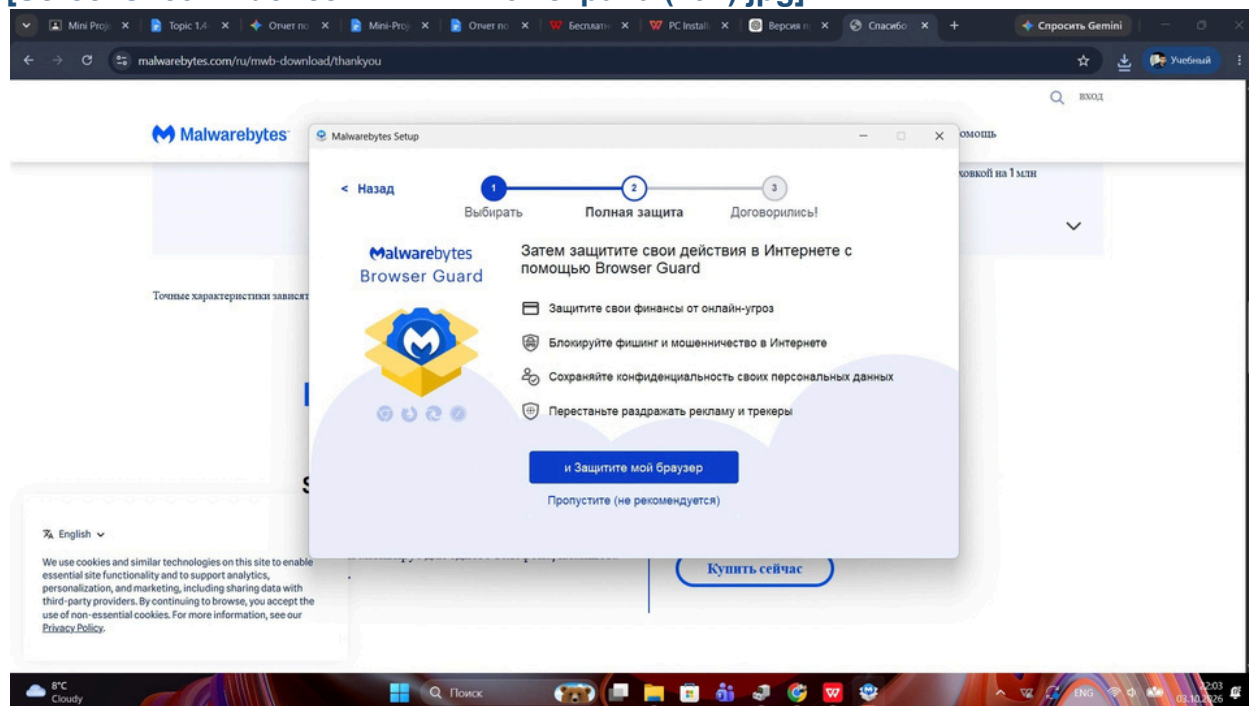
1. Downloaded the official setup file directly from the Malwarebytes website (malwarebytes.com).
2. Executed MBSetup.exe and selected "Personal Devices" (Для меня и моего дома) as the protection type.
3. Reviewed the optional Malwarebytes Browser Guard recommendation for web threat blocking.
4. Allowed the automated installation wizard to extract binaries, register real-time protection drivers, and complete setup.
5. Launched Malwarebytes, verified initial database updates, and prepared the system for a full threat scan.

[Screenshot Evidence 1 — Снимок экрана (193).jpg]



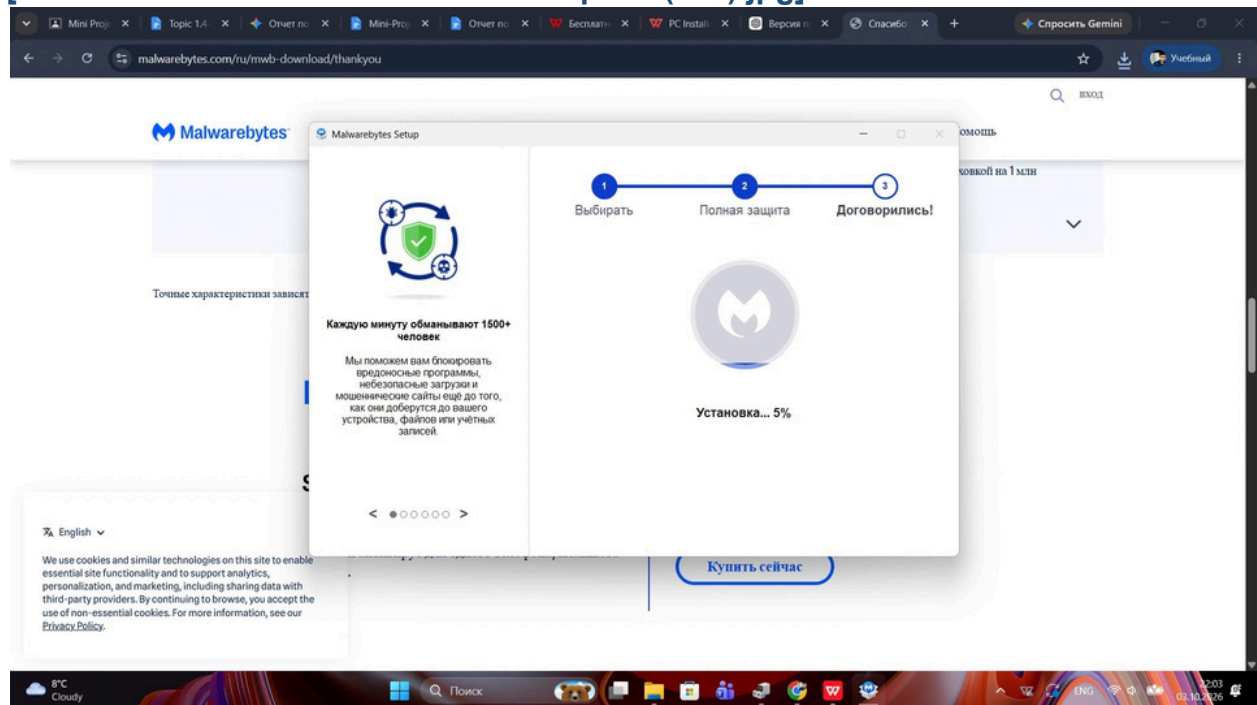
Description: Launching Malwarebytes Setup Wizard. Step 1: Selecting device usage type — 'Personal Devices' (Для меня и моего дома) selected.

[Screenshot Evidence 2 — Снимок экрана (194).jpg]



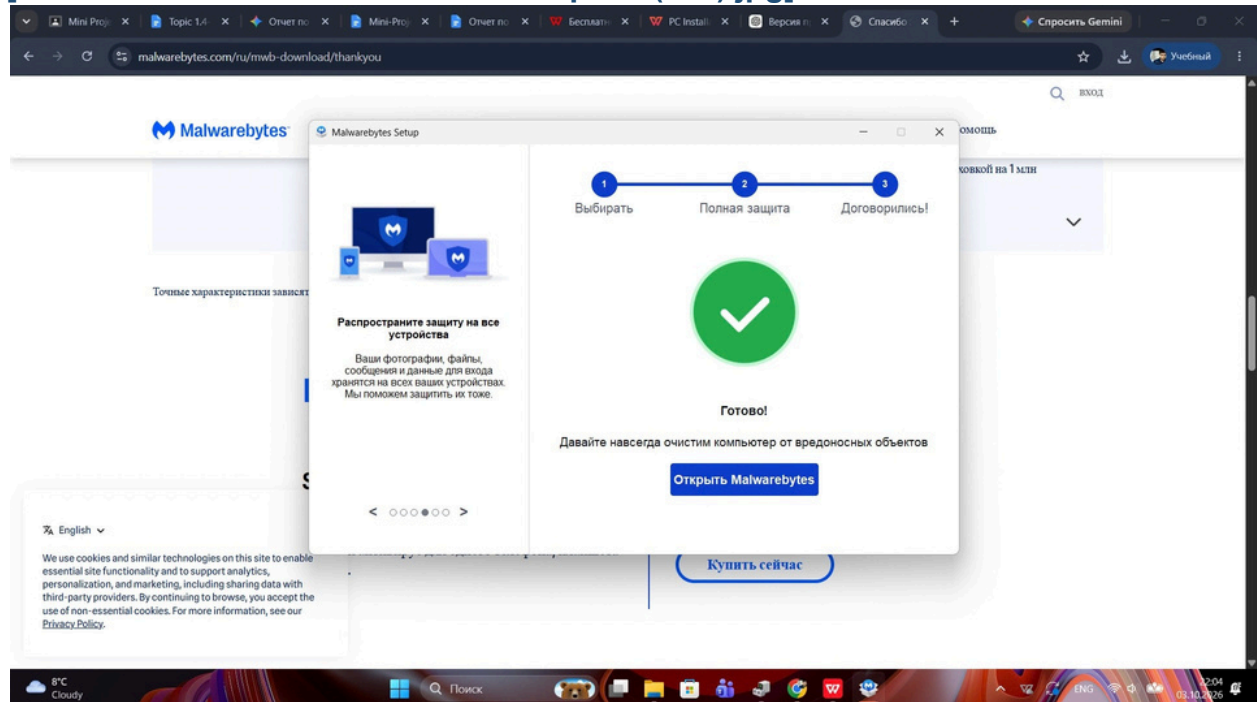
Description: Malwarebytes Setup Wizard Step 2. Recommendation to install Malwarebytes Browser Guard for online privacy, web protection, and ad blocking.

[Screenshot Evidence 3 — Снимок экрана (195).jpg]



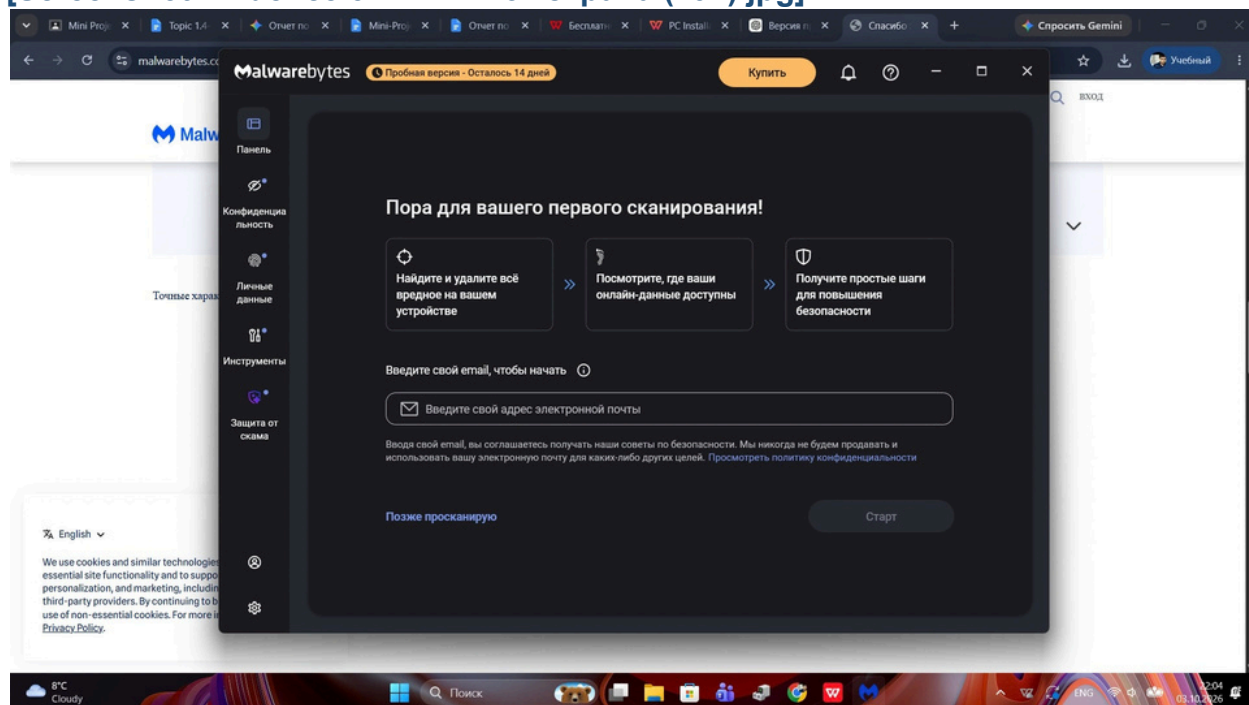
Description: Automated installation progress of Malwarebytes ("Установка... 5%"). Program files and background services being written to system directory.

[Screenshot Evidence 4 — Снимок экрана (196).jpg]



Description: Successful installation completion screen ("Готово! / Ready!"). Option available to launch Malwarebytes application.

[Screenshot Evidence 5 — Снимок экрана (197).jpg]



Description: First launch of Malwarebytes main interface ('Time for your first scan!'). 14-day Premium Trial active, real-time security shields operational.

3.3 Task 10: Antivirus Settings and Configuration

Following installation, the core protection modules and operational settings in Malwarebytes were verified and recorded as follows:

Setting / Feature	Selected Configuration	Operational Purpose
Real-Time Protection	Enabled (Web, Malware, Ransomware, Exploit Protection active)	Monitors file system and memory constantly to block threats before execution
Automatic Updates	Enabled (Automatically checks for update definitions every 1 hour)	Ensures threat signature database remains up to date against zero-day malware

Scheduled Scanning	Configured (Daily Threat Scan scheduled when system is idle)	Performs routine background system scans automatically without user intervention
Web Protection	Active (Web Protection engine + Malwarebytes Browser Guard extension)	Blocks malicious URLs, phishing domains, and malicious web scripts in real time
Quarantine Settings	Automatic Isolation (Move threats to Quarantine instantly)	Isolates detected threats automatically to prevent execution while preserving files for review
Notifications	Desktop Pop-ups Enabled for blocked threats and scan summaries	Provides immediate visual feedback when a threat or malicious item is intercepted

Scan Execution & Results: Executed initial Threat Scan on system volume. 0 malware or suspicious items detected; system security confirmed clean.